

Decision Control Structures in C (if-else, switch, ternary)

> **Definition to Remember**

1. Concept Overview & Fundamentals

Decision Control Structures in C (if-else, switch, ternary) forms a core cornerstone of computer systems programming, hardware-level software engineering, and competitive problem solving.

In the C execution model:

2. Technical Architecture & Memory Mechanics

Memory Layout & Storage Organization

```text

### Core Architecture Breakdown

| Component / Concept | Technical Specification | Operational Characteristics | Performance / Complexity |

| :--- | :

---

3. Core Syntax & Implementation Patterns

Essential Idioms & Header Declarations

```
```c
// Core standard implementation pattern
// Core logic execution
return status;
```
#include
int mai
printf("
}
```

---

4. Real-World Applications & System Engineering

- **Operating System Kernels:** Linux, UNIX, and Windows NT kernels utilize C for drivers, process scheduling, and memory paging.

- **E**

---

5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: AbsoluteTO

**Problem Description & Objective:**

**\*\*C Code Implementation:\*\***

```
#include <stdio.h>
```

```
int num, absolute;
```

```
printf("Enter number: ");
```

```
absolute = (num < 0) ? -num : num;
```

```
printf("Absolute value is %d\n", absolute);
```

```
return 0;
```

```

```

## **Problem 2: CategorizeAgeGroup**

**\*\*Problem Description & Objective:\*\***

**\*\*C Code Implementation:\*\***

Child -> below 13

```
int main() {
```

```
int age;
```

```
printf("Enter age: ");
```

```
scanf("%d", &age);
```

```
if (age <= 13) {
```

```
 printf("Child\n");
```

```
} else if (age <= 20) {
```

```
 printf("Teenager\n");
```

```
} else if (age <= 60) {
```

```
 printf("Adult\n");
```

```
} else {
```

```
 printf("Senior Citizen\n");
```

```
}
```

```
return 0;
```

```
}
```

```

```

### Problem 3: GradeCalculator

**\*\*Problem Description & Objective:\*\***

Implement

**\*\*C Code Implementation:\*\***

```
```c
```

A -> above 90%

B -> al

```
#include <stdio.h>
```

```
int main
```

```
float marks;

printf("Enter your marks: ");
scanf("%f", &marks);

if (marks >= 90) {
    printf("Grade A\n");
} else if (marks >= 75) {
    printf("Grade B\n");
} else if (marks >= 60) {
    printf("Grade C\n");
} else if (marks >= 30) {
    printf("Grade D\n");
} else {
    printf("You got FAIL!\n");
}

return 0;
}
```

Problem 4: GreatestNumber

****Problem Description & Objective:****

****C Code Implementation:****

```
#include<stdio.h>

int main()
{
    int num1,num2,num3;

    printf("Enter 1st number: ");
    scanf("%d", &num1);
    printf("Enter 2nd number: ");
    scanf("%d", &num2);
    printf("Enter 3rd number: ");
    scanf("%d", &num3);
}
```

```
if(num1 >= num2 && num1 >= num3){
```

```
printf("
```

```
return 0;
```

```
}
```

```
---
```

Problem 5: LeapYear

****Problem Description & Objective:****

Create
divisib

****C Code Implementation:****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int year;
```

```
printf("Enter year: ");
```

```
scanf("
```

```
if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
```

```
printf("%d is a Leap Year\n", year);
```

```
} else {
```

```
printf("%d is NOT a Leap Year\n", year);
```

```
}
```

```
return 0;
```

```
}
```

```

```

### Problem 6: MinimumNumberTO

**\*\*Problem Description & Objective:\*\***

Create

**\*\*C Code Implementation:\*\***

```
```c
```

```
#include<stdio.h>
```

```
int mai
```

```
int num1 , num2;
```

```
printf("Enter 1st number: ");
```

```
scanf(
```

```
(num1 < num2)?printf("1st is Minimum"):printf("2nd is Minimum");
```

```
return 0;
```

```
}
```

```
---
```

Problem 7: NumberChecker

****Problem Description & Objective:****

Create

****C Code Implementation:****

```
```c
```

```
#include<stdio.h>
```

```
int mai
```

```
int num;
```

```
printf("\nEnter Number: ");
```

```
if(num == 0){
```

```
return 0;
```

```

```

### Problem 8: OddEven

**\*\*Problem Description & Objective:\*\***

**\*\*C Code Implementation:\*\***

```
#include<stdio.h>
```

```
int num;
```

```
if(num == 0){
```

```
return 0;
```

```
}
```

```

```

### Problem 9: OddEvenTO

**\*\*Problem Description & Objective:\*\***

Create

**\*\*C Code Implementation:\*\***

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int num;
```

```
printf("Enter number: ");
```

```
scanf("%d", &num);
```

```
(num % 2 == 0) ? printf("Even\n") : printf("Odd\n");
```

```
return 0;
```

```
}
```

```
---
```

Problem 10: PrintMonthSwitch

****Problem Description & Objective:****

Create

****C Code Implementation:****

```
```c
```

```
#include <stdio.h>
```

```
int main() {
```

```
/*
```

```
int month;

printf("Enter month number (1-12): ");

scanf("%d", &month);

switch(month) {

case 1:

printf("January\n");

case 2:

printf("February\n");

case 3:

printf("March\n");

case 4:

printf("April\n");

case 5:

printf("May\n");

case 6:

printf("June\n");

case 7:

printf("July\n");

case 8:

printf("August\n");

case 9:

printf("September\n");

case 10:

printf("October\n");

case 11:

printf("November\n");

case 12:

printf("December\n");
```

default:

printf("

return 0;

}

---

### Problem 11: ScoreCategoryTO

**\*\*Problem Description & Objective:\*\***

Create  
scores

**\*\*C Code Implementation:\*\***

```c

#include <stdio.h>

int main() {

/*

int score;

printf("Enter student score: ");

scanf("

(score > 80)

? print

return 0;

}

Problem 12: SimpleCalculatorSwitch

****Problem Description & Objective:****

Create
subtra

****C Code Implementation:****

```
#include <stdio.h>
```

```
int main() {
```

```
int num1, num2;
```

```
printf("Enter first number: ");
```

```
printf("Enter operator (+, -, *, /): ");
```

```
printf("Enter second number: ");
```

```
switch(op) {
```

```
case '+':
```

```
case '-':
```

```
case '*':
```

```
case '/':
```

```
default:
```

```
return 0;
```

```
---
```

```
```c
```

```
/*
```

```
char o
```

```
scanf("
```

```
scanf("
```

```
scanf("
```

```
printf("
```

```
printf("
```

```
printf("
```

```
printf("
```

```
printf("
```

```
}
```

## 6. Exam Scoring Points & Memory Revision

> **Must-Write Points**

> 1. if-

> **Quick Recall**

> `Boo